

Infrared Spectroscopy: A Written Lab Report

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Abstract

The purpose of this experiment was to uncover the functional groups present in several organic compounds through infrared spectroscopy and vibration of covalent bonds. In general, we utilized a spectrometer to gather an infrared, transmittance, graphical spectrum of several organic compounds (carvone, limonene, p-cymene, citral, citronellol, and menthone). In my experiment, I determined all the functional groups that were present in my several samples. Finally, it was concluded that all the prevalent functional groups that exist within each of the compounds were discovered through my infrared spectroscopy experiment.

Introduction

This report encloses the importance of infrared spectroscopy in chemistry when determining the functional groups that are present within a compound. During the experiment, infrared light is shot onto a compound in which the absorbed light will vibrate the bonds. The vibrations are quantified as a change in dipole moments across the functional group. In general, the bonds will vibrate at higher frequencies depending on the size of the atoms and the strength of the bond – larger atoms will vibrate at lower frequencies while stronger bonds of the same atom will vibrate at higher frequencies. During the spectroscopy interpretation, the higher the wavenumber, the greater the frequency, with peaks appearing based on the transmittance of the infrared light (portions that were not absorbed). It is important to note that a broad peak could be caused by intermolecular interactions interfering, while a strong versus weak peak will depend on the intensity of the dipole moment created during the experiment. Finally, it is vital to understand that not all functional groups will be seen on an infrared spectroscopy graph, only ones that experience a dipole moment change.

The purpose of this experiment is to identify the functional groups that are present in carvone, limonene, p-cymene, citral, citronellol, and menthone. In short, during the experiment, I utilized an infrared spectrometer to cast light onto several compounds and determine their functional groups present based on the location of peaks represented on printed, computerized graphs.

For this experiment, I expect to receive appropriate graphical data representing the common peak values for several functional groups that are present within our six organic compounds. I hypothesize that each infrared spectroscopy for all six compounds will reveal their retrospective functional groups without much deviation from calculated values.

Finally, it is important to note that this report is outlined into two parts. Part I demonstrates how to calculate frequency values and their relationships while part II demonstrates the lab experiment.

Procedure

The procedure of this experiment entails three portions, and they are as follows:

Materials

A list of utilized chemicals include:

- Carvone
- Citral
- Citronellol
- Menthol
- p-Cymene
- Limonene

A list of utilized equipment include:

- IR Spectrometer

Equations

- $\nu = 4.120\left(\frac{K}{\mu}\right)^{1/2}$
 - $K = 5 \times 10^5$ (Single Bond)
 - $K = 10 \times 10^5$ (Double Bond)
 - $K = 15 \times 10^5$ (Triple Bond)
- $\mu = \frac{m_1 \times m_2}{m_1 + m_2}$

Methods

For part I, four organic compounds (benzene, benzene-d₆, isopropanol, and acetone) were utilized to calculate frequency values of certain functional groups and compare them with experimental values.

For part II, six terpenes (carvone, menthone, citronellol, citral, limonene, and p-cymene) were utilized to take an infrared spectroscopy – all liquids. For this procedure, calibrate the spectrometer, clean the machine crystal, then place a small portion of the liquid sample on the center crystal. Additionally, the HATR (Horizontal Attenuated Total Reflectance) device installed within the spectrometer will handle the infrared mechanisms that are dealt within this experiment, so simply begin the diagnostic and print the graphical representation. Finally, clean the crystal with isopropanol and a Kleenex and repeat the procedure for each liquid.

Results

Part I:

Figure 1: Benzene Infrared Spectroscopy – Graph

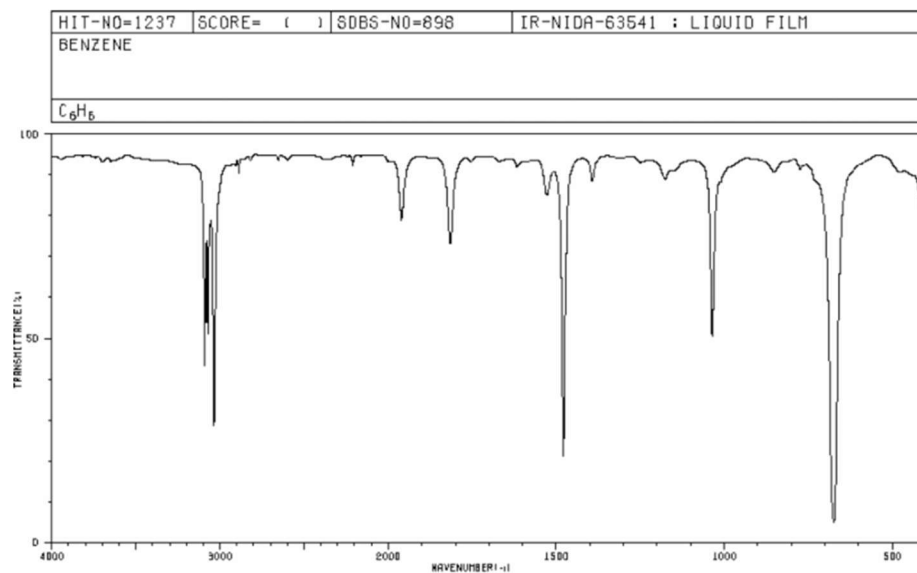


Figure 2: Benzene- d_6 Infrared Spectroscopy – Graph

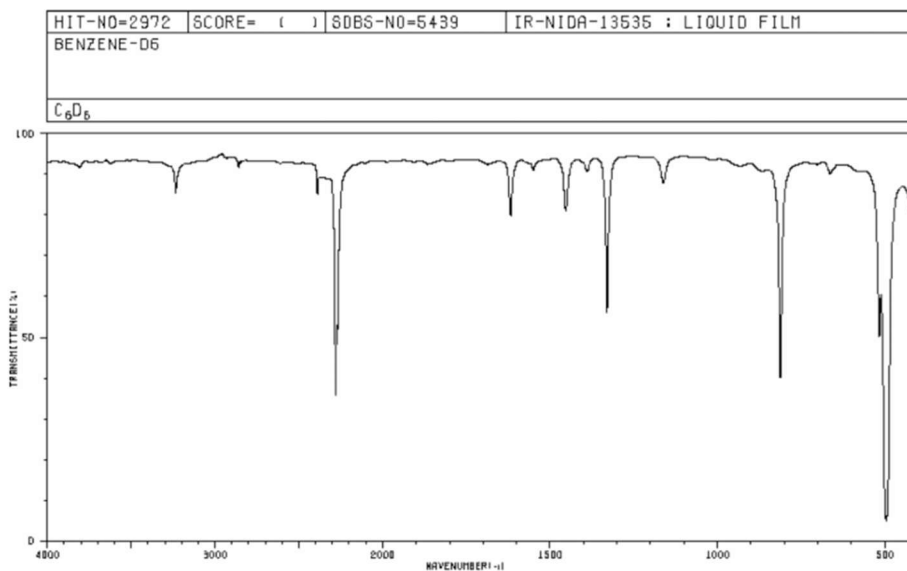


Figure 3: Isopropanol Infrared Spectroscopy – Graph

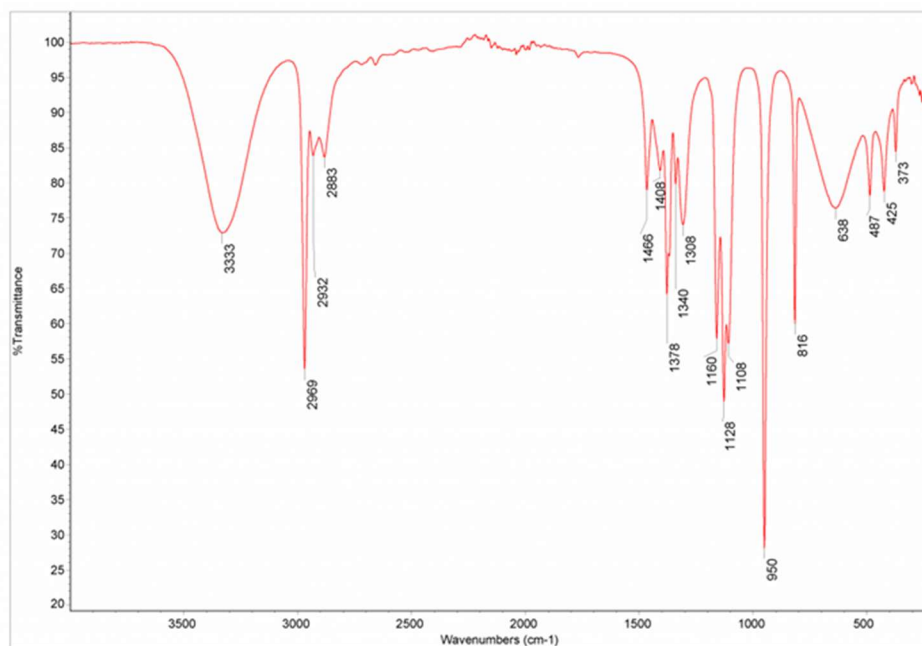


Figure 4: Isopropanol Infrared Spectroscopy – Graph

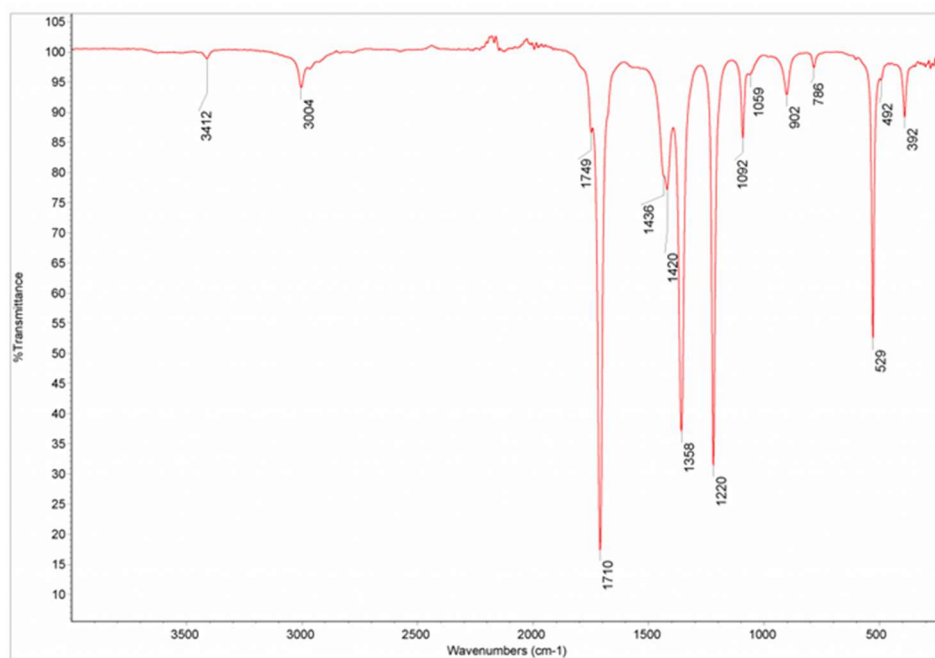


Table 1: Infrared Spectroscopy of Different Bonds

Bond Type	Calculated Wavenumber	Experimental Wavenumber
C-H bond in Benzene	3032 cm ⁻¹	3056 cm ⁻¹
C-D bond in Benzene-d ₆	2225 cm ⁻¹	2270 cm ⁻¹
C-O bond in Isopropanol	1112 cm ⁻¹	1128 cm ⁻¹
C=O bond in Acetone	1573 cm ⁻¹	1436 cm ⁻¹

Calculation 1: Example Math – C-H bond in Benzene

$$\triangleright \nu = 4.120 \left(\frac{K}{\mu} \right)^{1/2}$$

$$\triangleright K = 5 \times 10^5$$

$$\triangleright \mu = \frac{m_1 \times m_2}{m_1 + m_2}$$

$$\triangleright m_1 = 12 \text{ amu}$$

$$\triangleright m_2 = 1 \text{ amu}$$

$$\bullet \nu = 4.120 * \left(\frac{5 \times 10^5}{\frac{12 \times 1}{12 + 1}} \right)^{1/2}$$

$$\bullet \nu = 3032 \text{ cm}^{-1}$$

Part II:

Figure 5: Carvone Infrared Spectroscopy – Graph

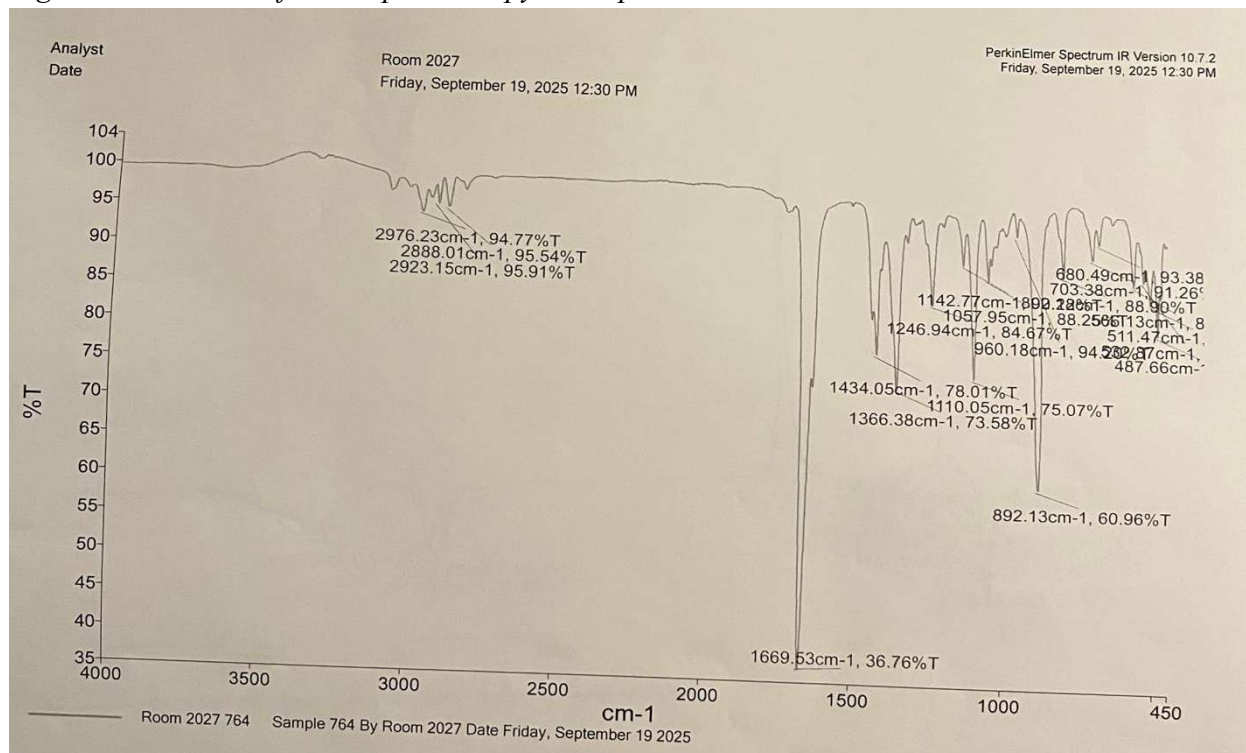


Table 2: Carvone Infrared Spectroscopy – Functional Groups

Major IR Frequencies	Assignments
2923 cm ⁻¹	C-H
2888 cm ⁻¹	C-H
2976 cm ⁻¹	C-H
1434 cm ⁻¹	C=C
1366 cm ⁻¹	C=C
1669 cm ⁻¹	C=O

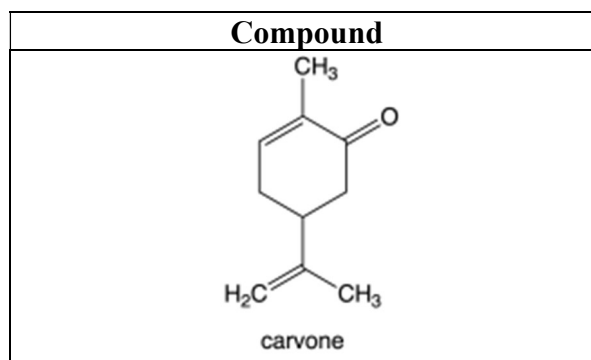


Figure 6: Limonene Infrared Spectroscopy – Graph

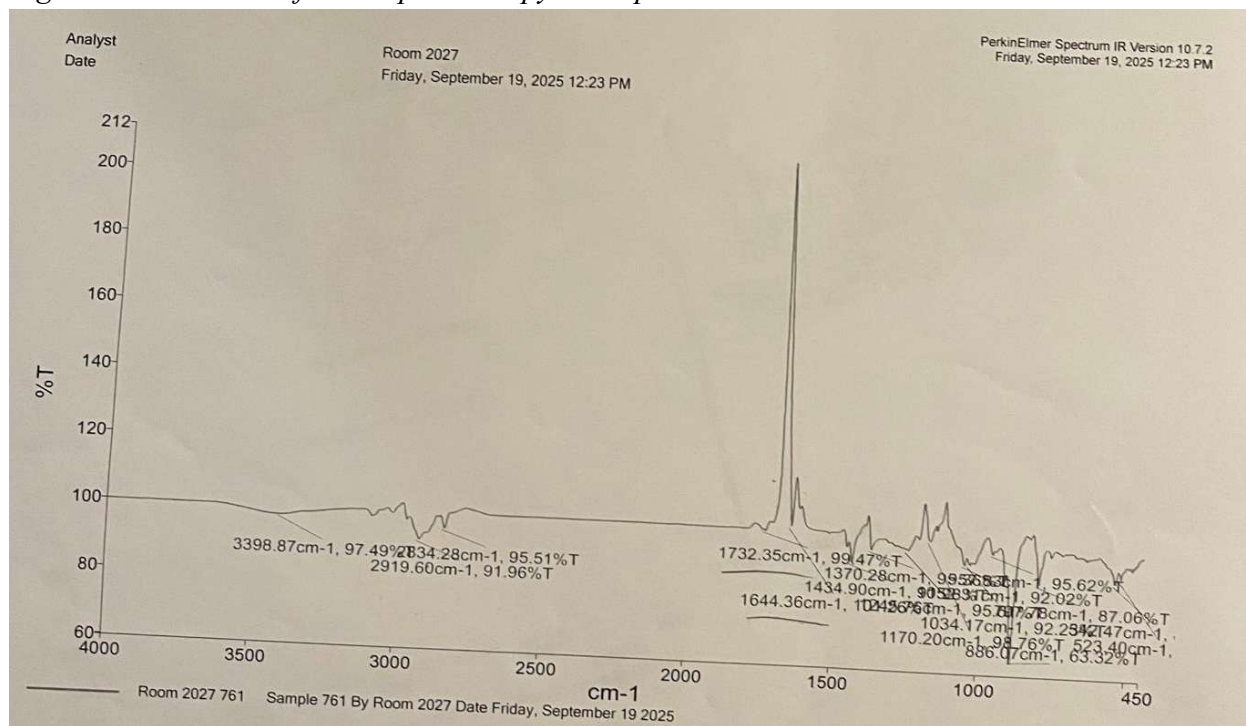


Table 3: Limonene Infrared Spectroscopy – Functional Groups

Major IR Frequencies	Assignments
2932 cm ⁻¹	C-H
2912 cm ⁻¹	C-H
2854 cm ⁻¹	C-H
1732 cm ⁻¹	C=C
1644 cm ⁻¹	C=C

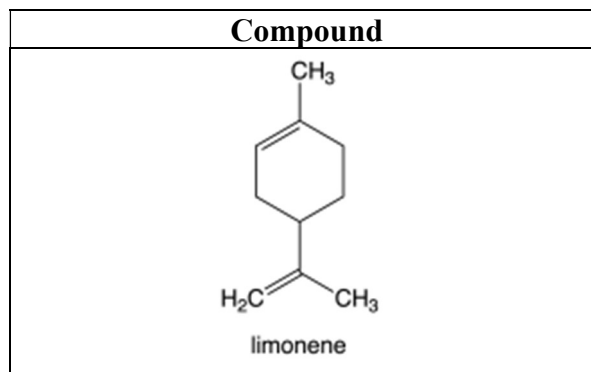
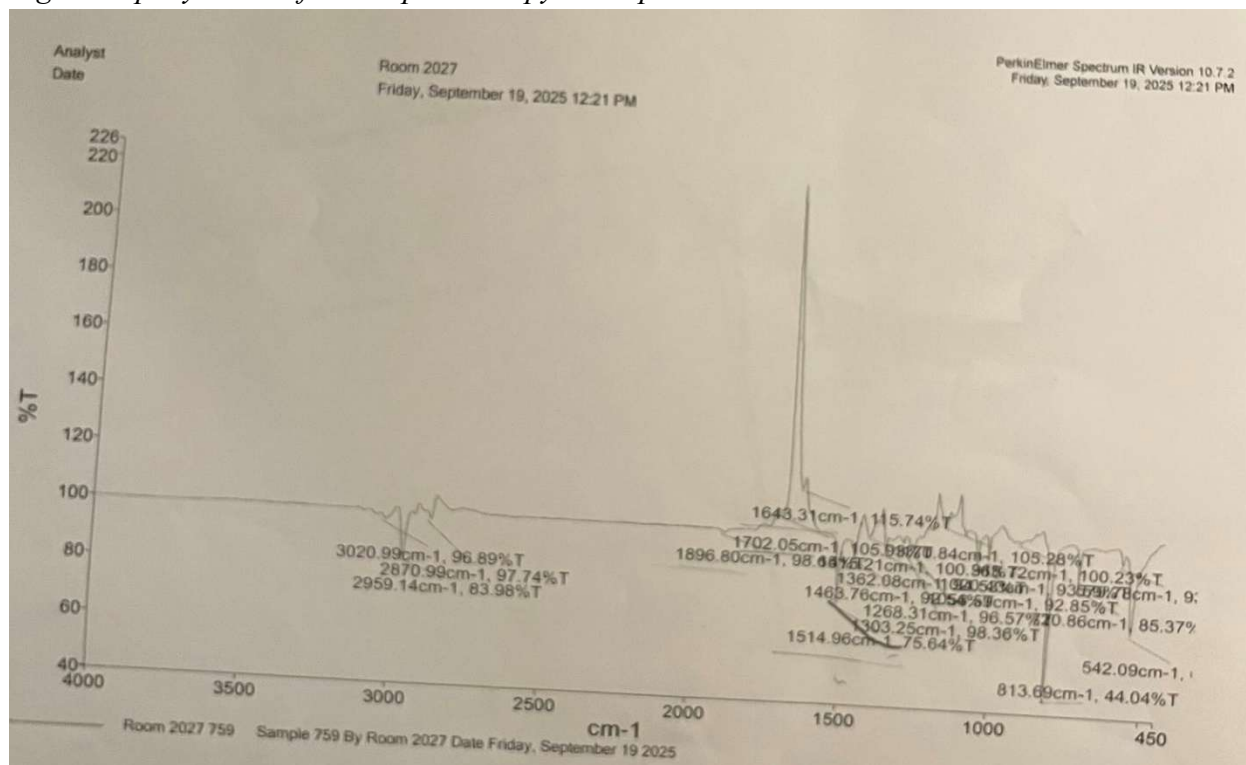


Figure 7: *p*-Cymene Infrared Spectroscopy – GraphTable 4: *p*-Cymene Infrared Spectroscopy – Functional Groups

Major IR Frequencies	Assignments
3020 cm ⁻¹	C-H
2870 cm ⁻¹	C-H
2959 cm ⁻¹	C-H
1702 cm ⁻¹	C=C
1643 cm ⁻¹	C=C
1514 cm ⁻¹	C=C

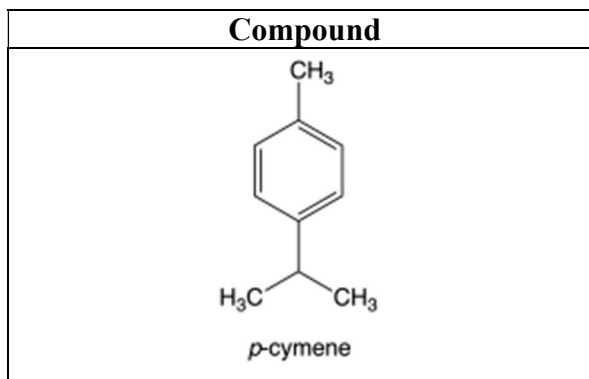


Figure 8: Citral Infrared Spectroscopy – Graph

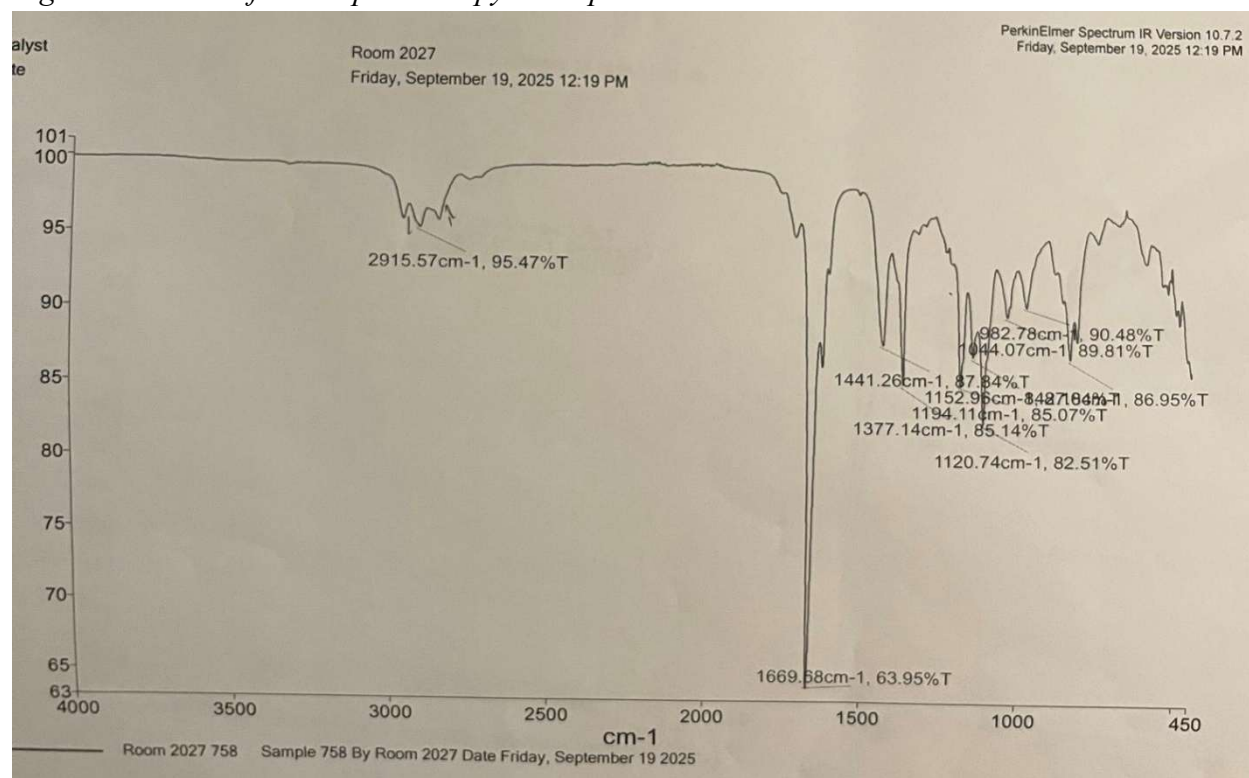


Table 5: Citral Infrared Spectroscopy – Functional Groups

Major IR Frequencies	Assignments
2915 cm ⁻¹	C-H
2915 cm ⁻¹	C-H
2915 cm ⁻¹	C-H
1669 cm ⁻¹	C=O
1441 cm ⁻¹	C=C

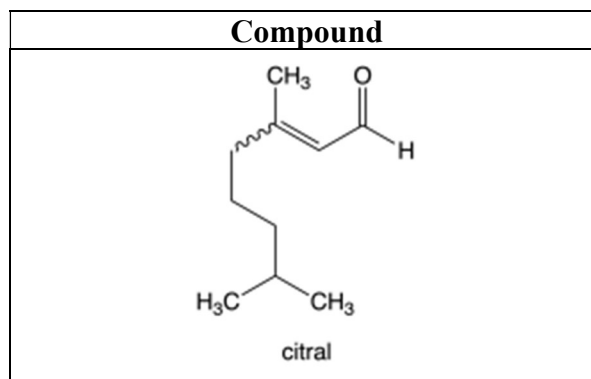


Figure 9: Citronellol Infrared Spectroscopy – Graph

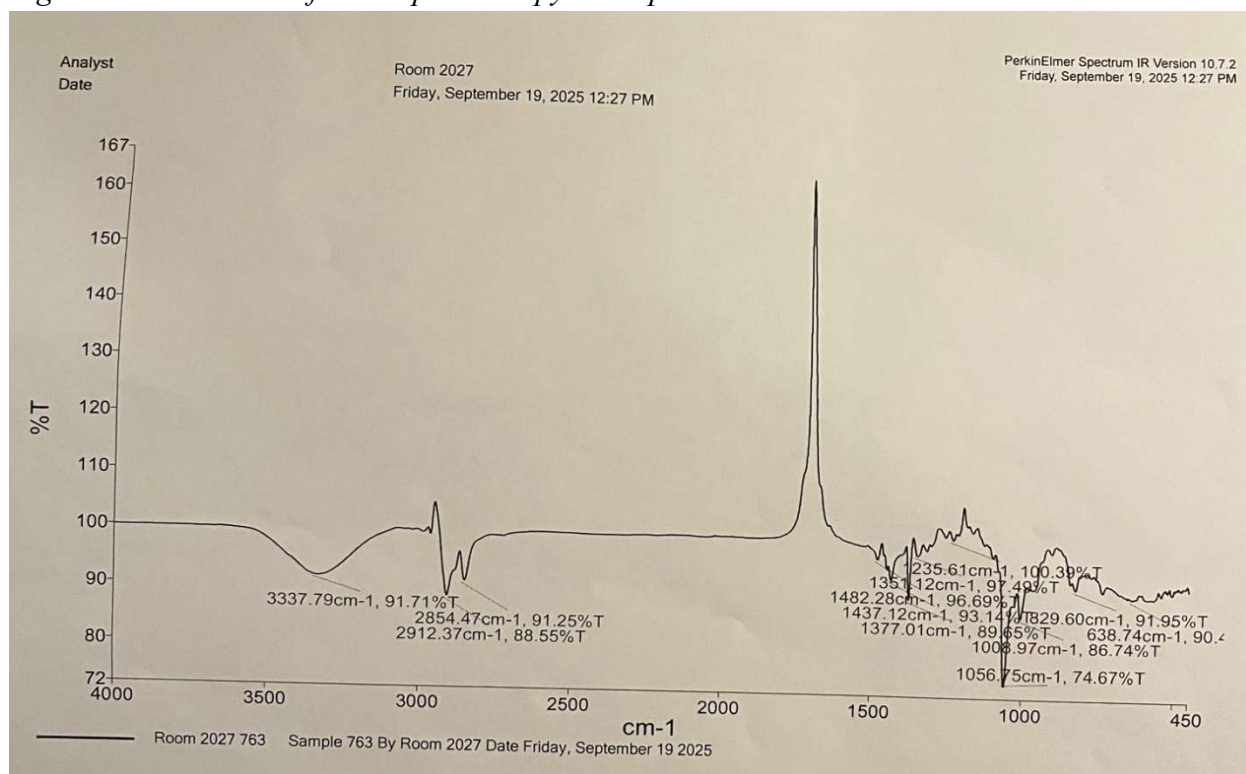


Table 6: Citronellol Infrared Spectroscopy – Functional Groups

Major IR Frequencies	Assignments
3337 cm ⁻¹	O-H
2912 cm ⁻¹	C-H
2912 cm ⁻¹	C-H
2854 cm ⁻¹	C-H
1482 cm ⁻¹	C=C

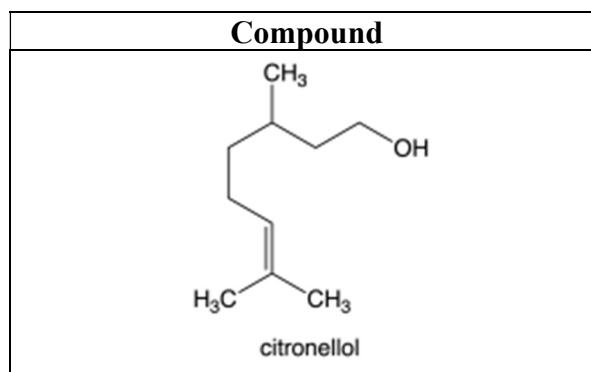


Figure 10: Menthone Infrared Spectroscopy – Graph

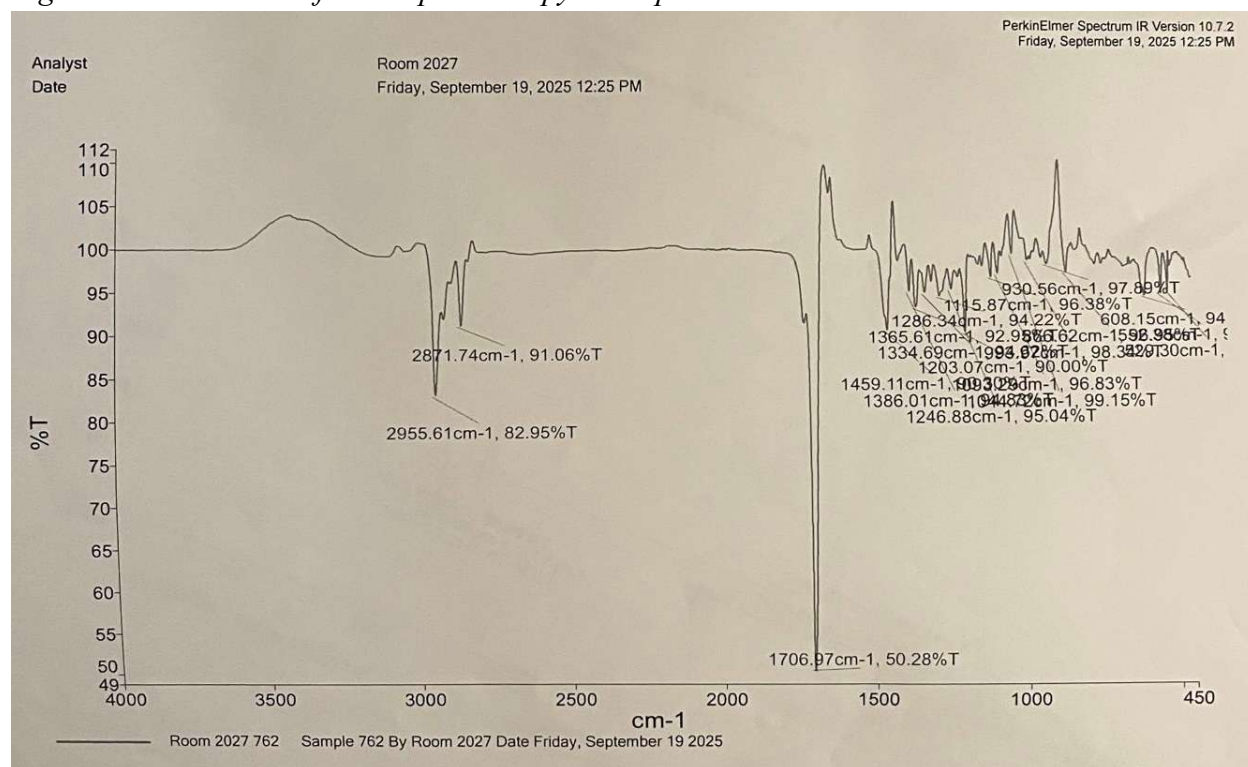
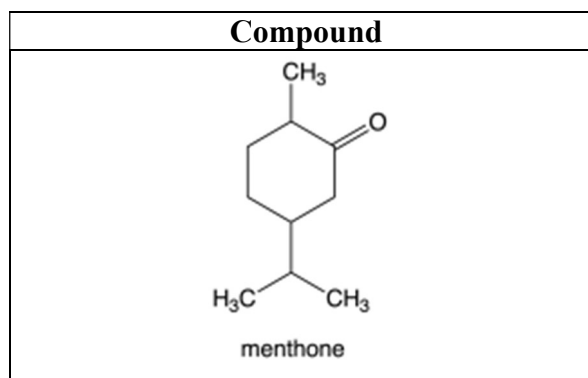


Table 7: Menthone Infrared Spectroscopy – Functional Groups

Major IR Frequencies	Assignments
2955 cm ⁻¹	C-H
2871 cm ⁻¹	C-H
2871 cm ⁻¹	C-H
1706 cm ⁻¹	C=O



Discussion

Part I:

This section demonstrates how to calculate the frequency for a functional group bond (*Calculation 1*) and the relationship between calculated and experimental wavenumber values. When evaluating the discrepancies between values in Benzene and Benzene-d₆ (*Figure 1, Figure 2, & Table 1*), it is important to note that the absorption difference is due to the “D” in Benzene-d₆ having a larger atomic mass, thus, a lower wavenumber frequency. Additionally, the difference between the frequency values in Isopropanol and Acetone (*Figure 3, Figure 4, & Table 1*) is due to the double bond functional group that acetone has. This is because a double bond of the same atomic components will yield a greater frequency due to a larger change in dipole moments when vibrating due to infrared interference – stronger bond strength.

Finally, the reason the experimental and calculated values differ is due to the chemical environment of the experimental and relevant bond strength while the calculated only measures the strength of the covalent bond.

Part II:

This section analyzes the results of the experiment. My data revealed the accurate functional groups of the provided terpenes through infrared spectroscopy. Carvone correctly displayed the appropriate C-H, C=C, and C=O bonds present in its ring structure (*Figure 5 & Table 2*). Additionally, Limonene and p-Cymene accurately represented the C-H and C=C functional groups that exist within its organic compound (*Figure 6, Table 3, Figure 7, Table 4*). Continuing, Citral's data effectively reveals the C-H, C=O, and C=C bonds that are present within its skeleton structure (*Figure 8 & Table 5*). Furthermore, Citronellol correctly displayed the appropriate O-H, C-H, and C=C functional groups present in its skeleton structure (*Figure 9 & Table 6*). Finally, Menthone accurately represented the C-H and C=O bonds that were present within its ring structure (*Figure 10 & Table 7*). Overall, all the functional groups present were accurately revealed.

Regarding my expected results, this experiment agrees with my hypothesis. All the compounds were accurately depicted, revealing all their relevant functional groups present within their organic structure.

All results were expected, but to account for the minor discrepancies within the infrared spectrum values, it could be from an assortment of reasons. For one, the center crystal of the spectrometer could have been slightly contaminated with residue remaining on its surface area. Additionally, some of the samples may have been slightly impure, resulting in a mixed measure value. Finally, intermolecular interactions and environment could have been factors to slightly skew the results of the experiment. Overall, the data was fairly accurate to known values.

Conclusion

In summary, this experiment has succeeded in adequately representing the correct functional groups present in several organic compounds (carvone, limonene, p-cymene, citral, citronellol, and menthone). Additionally, this experiment supports my original hypothesis that it will report all the functional groups present within each organic material without much deviation from the actual value. Furthermore, any slight discrepancies from the calculated, actual value of each of the functional groups may be from human error, intermolecular bond interactions, and the chemical environment, but none of these were significant enough to skew the values unrealistically. Finally, for the future, I would suggest that more careful craftsmanship and an ideal environment would be utilized to execute this experiment with the most minimal error possible.